

## DAY 4 SQL – Query Analysis & Result Explanation

### Concepts & Topics Used (Explained)

- CREATE TABLE: Creates a new table in the database; result is an empty table structure.
- INSERT INTO: Adds new rows into a table; result is stored records.
- SELECT: Retrieves data from one or more tables as query output.
- WHERE: Filters rows so only matching records appear in result.
- PRIMARY KEY: Ensures each row is uniquely identifiable.
- FOREIGN KEY: Links child table data to parent table records.
- AUTO\_INCREMENT: Automatically generates increasing numeric values.
- UNIQUE: Prevents duplicate values in a column.
- COMMIT / AUTOCOMMIT: Saves transaction changes permanently in database.

#### ◆ UNION

UNION combines the result of two or more SELECT queries.

It removes duplicate rows from the final result set.

Each SELECT must have the same number of columns and compatible data types.

It is commonly used when unique combined data is required.

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#### ◆ UNION ALL

UNION ALL combines the result of multiple SELECT queries.

It does not remove duplicate rows from the output.

It is faster than UNION because no duplicate checking is done.

It is used when all records, including duplicates, are needed.

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#### ◆ INDEX

An INDEX is used to improve the speed of data retrieval operations. It helps the database find rows faster without scanning the full table. Indexes are commonly used on columns in WHERE and JOIN conditions. Too many indexes can slow down INSERT and UPDATE operations.

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#### ◆ VIEW

A VIEW is a virtual table created using a SELECT query. It does not store data but displays data from underlying tables. Views improve query readability and data security. They are useful for simplifying complex SQL queries.